

Technology Stack

Date	01 July 2026
Team ID	
Project Name	HDI Predictor using Machine Learning and Flask
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	<i>HTML5, CSS3, JavaScript (ES6+), Tailwind CSS (via CDN), Chart.js</i>	<i>Selected for constructing a highly responsive, mobile-friendly interface. Tailwind CSS allows rapid utility-based styling, while Chart.js enables dynamic client-side rendering of radar and bar charts to visually compare normalized indicators (life expectancy, schooling, income).</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python 3.9+, Flask Web Framework (v3.x), Scikit-Learn</i>	<i>Python is the industry standard for ML. Flask is a lightweight micro-framework chosen for its minimal setup, ease of building RESTful API endpoints, and direct compatibility with Scikit-Learn prediction modules.</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>Flat-file CSV storage ('HDI_Cleaned.csv'), serialized pickle model ('HDI.pkl')</i>	<i>A flat-file system is highly efficient here. The database of 190+ countries is maintained in a cleaned CSV file for autocomplete features, and the Linear Regression ML model is serialized via 'pickle' for instant, sub-millisecond loading on server startup without database overhead.mapping)]</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Docker, Render (or AWS Elastic Beanstalk)</i>	<i>Docker containers guarantee consistency across local development and production environments. Render or AWS Elastic Beanstalk offers cost-effective, auto-scaling deployment environments for Python/Flask applications.</i>
5	Version Control & CI/CD	<i>Git, GitHub, GitHub Actions</i>	<i>it and GitHub ensure collaborative version control and history tracking. GitHub Actions automates unit tests, dependency audits, and syntax formatting checks on commits.</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Google Fonts API, CDN Deliveries (Tailwind/Chart.js)</i>	<i>Google Fonts supplies clean modern typography (e.g., *Plus Jakarta Sans*), while CDN deliveries render CSS styling and chart behaviors instantly on the browser side without inflating the local application footprint.</i>